

HAFIZ MUHAMMAD ADNAN TARIQ

Bioinformatics Researcher | Multi-Omics Analysis | AI & Reproducible Pipelines

📍 Islamabad 44000, Pakistan 📞 (+92) 332-6400747

✉️ tadnan329@gmail.com

🌐 [LinkedIn](#) | [GitHub](#) | [ORCID](#)

Bioinformatician and computational biology researcher with research experience in metagenomics, microbiome-host interactions, machine learning, and clinical genomics. Experienced in developing reproducible bioinformatics workflows for large-scale sequencing data analysis using Nextflow, Python, R, and Linux/HPC environments. Research interests include multi-omics integration, translational microbiome research, and AI-driven approaches for precision medicine.

RESEARCH INTERESTS

- Microbiome and host-microbiome interactions
- Multi-omics integration (genomics, metagenomics, metabolomics – learning focus)
- Machine learning for biological prediction models
- Reproducible and scalable bioinformatics workflows

INDEPENDENT PROJECT

Reproducible Clinical Genomics Pipeline (Nextflow, DSL2, Docker) 2026 - Present

- Designed and implemented a modular, reproducible pipeline using Nextflow (DSL2) and Docker containers
- Automated end-to-end processing of NGS data (QC → alignment → variant calling → annotation)
- Focused on reproducibility, scalability, and portability across computing environments
- Currently testing and optimizing performance for real-world datasets

RESEARCH EXPERIENCE

Identification of Key Taxonomic and Metabolic Players in the Gut Metagenome of T2D Patients, NUST University (MS-Thesis Project) 07/2023-08/2024

- Investigated taxonomic and functional signatures in the gut microbiome of Type 2 Diabetes patients
- Identified key microbial taxa and carbohydrate-active enzymes (CAZymes) associated with disease state
- Applied metagenomic pipelines (MetaPhlAn, HUMAnN), statistical modelling, and machine learning.

Machine learning identifies microbial biomarkers of environmental antibiotic exposure in zebrafish gut metagenomes, China 10/2023-06/2024

- Analyzed gut metagenomic data to identify microbial biomarkers of environmental antibiotic exposure.
- Applied machine learning models to detect antimicrobial resistance-associated patterns.

Gut Microbial Strains - Non-Small Cell Lung Cancer Treatment: Decoding the Connection, NUST University 08/2024-06/2025

- Developed statistical and machine learning models to associate microbial variants/strains with immunotherapy responses, achieving >90% prediction accuracy.
- Applied structural bioinformatics to assess functional impact of SNPs on protein stability and active-site interactions, contributing to biomarker discovery for personalized NSCLC treatment.

RESEARCH PUBLICATIONS

1. Tariq, H.M.A., Khan, N.Y., Manzoor, H., & Kayani, M.U.R. Exploring the Impact of Type 2 Diabetes and Glucose-Lowering Drugs on Gut Microbiome Dynamics. Springer Nature Discover Medicine. 2025
DOI: <https://doi.org/10.1007/s44337-025-00241-9>
2. Raziq, M.F., Tariq, H.M.A., Khan, N., Manzoor, H., Rafiq, M., Rasool, S., Kayani, M.U.R., & Huang, L. Prioritizing gut microbial SNPs linked to immunotherapy outcomes in NSCLC patients by integrative bioinformatics analysis. Journal of Translational Medicine. 2025
DOI: <https://doi.org/10.1186/s12967-025-06370-0>

PROFESSIONAL EXPERIENCE

- Geneticist & Bioinformatician, ZGL, Islamabad Pakistan. ([LinkedIn](#))** 07/2025-Present
- Perform large-scale analysis of WES/WGS data, including variant calling, annotation, and clinical interpretation
 - Lead bioinformatics workflows for NGS-based diagnostics across gene panels, exomes, and genomes
 - Develop a containerized Nextflow (DSL2) pipeline for reproducible clinical genomics analysis (in progress), integrating multiple tools from raw reads to annotated variants
 - Work within Linux/HPC environments to manage and process high-throughput sequencing datasets
 - Collaborate with clinicians and researchers to translate genomic findings into actionable insights.
- Machine Learning Trainee, Intelli Ware Tech, Islamabad, Pakistan. ([LinkedIn](#))** 09/2024-04/2025
- Hands-on experience with **Bio Python** for biological data analysis.
 - Designed and implemented machine learning algorithms for real-world datasets
 - Collaborated in a team-based environment, applying Python, NumPy, Pandas, and scikit-learn to develop ML pipelines.

COMPUTATIONAL & BIOINFORMATICS SKILLS

Omics & Microbial Data Analysis:

Metagenomics, microbial community profiling, NGS data analysis, variant calling & annotation

Programming & Workflow Development:

Python (pandas, scikit-learn), R (tidyverse), Bash, Nextflow (DSL2), Docker

Tools & Frameworks:

MetaPhlAn, HUMAnN, Kraken2, QIIME2, StrainPhlAn, FastQC, MultiQC

Statistical & Machine Learning:

Random Forest, XGBoost, Logistic Regression, cross-validation, feature selection

Computing Environment:

Linux, high-performance computing (HPC), reproducible workflow development

INTERSHIPS

- Research Intern Computational Biology Lab, National Centre for Bioinformatics – QAU Islamabad, Pakistan.** 2021-2021
- Conducted in silico analysis of bioactive compounds targeting disease-related pathways
 - Investigated RNA-binding protein inhibition as a therapeutic strategy for kidney disorders
- Internship Nishtar Medical University, Multan, Pakistan** 2019
- Assisted in sample preparation and testing using laboratory equipment.
 - Gained hands-on experience in biochemistry, microbiology, and virology laboratory techniques.

RESEARCH ENGAGEMENT & TRAINING

Facilitator – Hands-On NGS Workshop: Metagenomics for Health and Disease NUST – Islamabad, Pakistan	02/2025
Co-facilitated a two-day practical workshop focused on NGS and Metagenomics, attended by students and researchers from interdisciplinary backgrounds.	
2 Days Hands-on Workshop on Complete Data Analysis of Single-Cell RNA-Seq	02/2024
Performed end-to-end single-cell RNA-seq analysis of case-control samples.	

EDUCATION

MS Bioinformatics	09/2022-08/2024
National University of Sciences and Technology (NUST), Islamabad	
Thesis: Identification of Key Taxonomic and Metabolic Players in the Gut Metagenome of T2D Patients, NUST University	
BS Bioinformatics	09/2017-08/2021
Quaid-i-Azam University (QAU), Islamabad, Pakistan.	

LANGUAGES

URDU	Native Language	
ENGLISH	IELTS (B2)	05/2025

COURSES AND CERTIFICATES

Interpreting Genomic Variation: Overcoming Challenges in Diverse Populations	2025
Wellcome Connecting Science	
Introduction to Genomic Technologies, Coursera	2023
Johns Hopkins University.	
Python For Genomic Data Science, Coursera	2023
Johns Hopkins University.	

LEADERSHIP & SCIENTIFIC COMMUNITY

Lead Outreach	01/2024-12/2024
International Society for Computational Biology (ISCB), Pakistan	
Director Outreach	03/2022-03/2023
NUST Bazm – e – Pakistan, Islamabad, Pakistan	
Co – organizer: African Student Council Symposium	04/2025
ISCB – SC, Cape Town, South Africa.	

REFERENCES

Dr. Masood Ur Rehman Kayani (Supervisor)	Muhammad Mazhar Saeed,
Associate Professor,	Manager Admin & HR,
National University of Sciences & Technology (NUST),	Zarqa Genomics Lab (ZGL), Islamabad,
Islamabad, Pakistan.	mazhar.fin@zglgenomics.com
Email: m.kayani@sines.nust.edu.pk	
Telephone: 05190855744	